Homework 3

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Problem 1

Let $C_{n-1}(t)$ be the space of continuous piecewise linear functions on $[0, n-1]$ whose slope is constant on each interval $[i-1, i]$, $i = 1, \ldots, n-1$. Define $\mathcal{A} : C_{n-1}(t) \to C_{n-1}(t)$ by

$$\mathcal{A}[f(t)] = f(n-1-t).$$

(1) Prove that $\mathcal{A}$ is a linear transformation.

(2) Let $\epsilon_j(t) \in C_{n-1}(t)$, $j = 1, \ldots, n$, be the nodal basis satisfying $\epsilon_j(i-1) = \delta_{ij}$ for $i, j = 1, \ldots, n$. Determine the matrix representation of $\mathcal{A}$ with respect to this basis.

Solution

(1) Reflection $t \mapsto n-1-t$ maps grid intervals to grid intervals, preserving continuity and piecewise linearity. Thus the map is well-defined. For functions f, g and scalars a, b,

$$\begin{aligned}\mathcal{A}(af + bg)(t) &= af(n-1-t) + bg(n-1-t) \\ &= a(\mathcal{A}f)(t) + b(\mathcal{A}g)(t),\end{aligned}$$

so $\mathcal{A}$ is linear.

(2) At every node,

$$(\mathcal{A}\epsilon_j)(i-1) = \epsilon_j(n-i) = \delta_{n+1-i,j} = \delta_{i,n+1-j}.$$

Since nodal values uniquely determine a function, $\mathcal{A}\epsilon_j = \epsilon_{n+1-j}$. Hence

$$\boxed{[\mathcal{A}]_\epsilon = J_n = (\delta_{i,n+1-j})_{i,j=1}^n}, \qquad (J_n)_{ij} = \begin{cases} 1, & i+j = n+1, \\ 0, & i+j \neq n+1. \end{cases}$$

This is the reversal matrix, with ones on the antidiagonal and zeros elsewhere.

Problem 2

Let I_1 and I_2 be the polynomial ideals generated by

$$a_1(\lambda) = (\lambda+1)(\lambda+2)^2, \qquad a_2(\lambda) = (\lambda+2)(\lambda+3)(\lambda+5),$$

respectively. Compute $I_1 + I_2$ and $I_1 \cap I_2$.

Solution

Work in $\mathbb{F}[\lambda]$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Using $I_1 + I_2 = \langle \gcd(a_1, a_2) \rangle$ and $I_1 \cap I_2 = \langle \operatorname{lcm}(a_1, a_2) \rangle$, take the smaller and larger multiplicities of each factor, respectively. Thus

$$\boxed{I_1 + I_2 = \langle \lambda + 2 \rangle},$$

$$\boxed{I_1 \cap I_2 = \langle (\lambda+1)(\lambda+2)^2(\lambda+3)(\lambda+5) \rangle}.$$

Problem 3

Let I_1 and I_2 be the polynomial ideals generated by

$$a_1(\lambda) = \lambda^4 + \lambda^3 - \lambda - 1,$$
$$a_2(\lambda) = \lambda^4 + 2\lambda^3 + 3\lambda^2 + 2\lambda + 1,$$

respectively. Compute $I_1 + I_2$ and $I_1 \cap I_2$.

Solution

In the same coefficient field as Problem 2, set $q(\lambda) = \lambda^2 + \lambda + 1$. Factoring gives

$$a_1 = (\lambda + 1)(\lambda - 1)q = (\lambda^2 - 1)q, \qquad a_2 = q^2.$$

Since $q(1) = 3$ and $q(-1) = 1$, we have $\gcd(\lambda^2 - 1, q) = 1$. Therefore

$$\boxed{I_1 + I_2 = \langle \lambda^2 + \lambda + 1 \rangle},$$

$$\boxed{I_1 \cap I_2 = \langle (\lambda^2 - 1)(\lambda^2 + \lambda + 1)^2 \rangle}.$$